

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0513

Curran 138 kV
August 23, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0513. The Interconnection Customer proposes a 174.6 MW generating facility interconnecting at the Curran 138 kV substation in the West Virginia area. The request was received on August 27, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 289 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,105,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Pit 7 Tap / Ft Martin 138 KV
Contingency: Morgantown Energy Facility / Prides Corner 345 KV
Loading: 123.8% (Rating: 274 MVA)
DFAX: 8.8%
Pre-project loading: 118.1%
2. Facility: Ft Martin / Longview Power Plant 138 KV
Contingency: Prides Corner / Morgantown Energyfacility 345 KV L
Loading: 122.6% (Rating: 288 MVA)
DFAX: 8.8%
Pre-project loading: 117.2%
3. Facility: Ronco 500/345 kV Transformer
Contingency: Auburn Rd / Prides Corner 345 KV L
Loading: 100.0% (Rating: 613 MVA)
DFAX: 6.7%
Pre-project loading: 98.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pit 7 Tap / Ft Martin 138 KV	Rebuild line with higher-capacity conductor	\$5,810,000	7% (\$395,000)
Fort Martin / Longview Power Plant 138 KV	Rebuild line with higher-capacity conductor	\$564,000	7% (\$38,000)
Ronco 500-345 kV Auto	Replace transformer with higher-rated unit	\$23,415,000	3% (\$672,000)
	Total allocated to project		\$1,105,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.